Designing a Jurisdiction-Aware RegTech Tool

——AxisReg Analytics and JurisFair BNPL Credit Monitor

Task 2 - Values Audit

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1. Mission statement and company profile

AxisReg Analytics is a hypothetical growth-stage B2B RegTech company. Its mission is to make automated financial decisioning monitorable, explainable and governable across divergent regulatory regimes without pretending that legal judgement can be outsourced to software. The company has approximately 40 employees and specialises in model monitoring, fairness analytics, explainable AI, rule configuration, audit logging and governance reporting.

Its initial market is US-EU credit decisioning for BNPL and consumer credit. Its commercial aspiration is to expand later to the UK, Singapore, Hong Kong, Canada and Australia, but the current product deliberately remains narrow: US and EU credit scoring / AI governance.

2. Whose perspective the tool serves

The paying client is the lender ' s Chief Compliance Officer, Model Risk Committee and Legal / Regulatory Affairs team. This is a realistic commercial choice because these stakeholders have the budget, authority and operating need to buy a cross-jurisdiction compliance monitoring layer. The tool also indirectly serves consumers by detecting risk-adjusted approval disparities, weak denial-reason mapping and subgroup data drift.

The central tension is genuine: the business may prefer fewer false alarms and faster deployment, while consumers need fair access and meaningful explanations, and regulators need accountable governance. The design therefore uses amber/red/watch tiers and human review rather than turning every statistical alert into a legal conclusion.

3. Genuine risk detection versus documentation

The design choice that best reflects the values audit is Risk-Adjusted Approval Disparity (RAAD). RAAD is a substantive risk-detection metric because it compares approval outcomes within predicted-default-risk bands rather than relying on crude overall approval rates. A spreadsheet can show overall gaps, but it cannot reliably connect comparable-risk cohorts, denial reasons, rule versions, data drift and jurisdiction-specific governance actions.

The tool does produce documentation, including rule versions, model versions, denial reason mappings and governance logs. That documentation is necessary for auditability, but it is not allowed to substitute for real outcome monitoring. In EU mode, this is reinforced by treating PSI as a data-governance trigger rather than a fairness conclusion.

4. Who bears the cost if the tool gets it wrong

Error type	Specific harm	Stakeholder most affected	Design response
False positive	Unnecessary review cost, model launch delay and compliance burden.	Klarna business, compliance and model risk teams.	Use confidence intervals, amber/red tiers and human review.
False negative	A genuine disparity is missed, causing unfair denials and legal, regulatory or reputational exposure.	Consumers, especially the affected audit group.	Use |RAAD|, AIR screening, subgroup drift checks and sensitivity analysis.
Performance drift	Baseline model reliability deteriorates under stress or distribution shift.	Consumers and the firm.	Monitor PSI, subgroup PSI and reassess RAAD / FNR gaps.
Jurisdictional misconfiguration	US rules are applied to EU data or EU rules are applied to US data.	Both consumers and the firm.	Record jurisdiction, rule version, data version and model version; rerun under correct configuration.
Incomplete explanation mapping	Reasons do not reflect actual model or policy drivers.	Denied consumers and compliance teams.	Map reasons to both model features and residual policy-layer factors.